

Noel Sason

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EDUCATION

University of California, Berkeley — *B.A. Molecular Biology | B.A. Data Science* Aug 2024 – Dec 2027
GPA: 3.8 / 4.0 | **Relevant Coursework:** CS61B (Data Structures & Algorithms), Math 54 (Linear Algebra), Data 8 (Foundations of Data Science — Statistics & Probability), Data C88C (Computational Structures in Data Science)

EXPERIENCE

Research Affiliate — *Lawrence Berkeley National Laboratory* Jan 2026 – Present

- Enabled cross-cohort meta-analysis across 80+ disease–microbiome studies by building a DuckDB-backed pipeline over the KG-Microbe knowledge graph for SQL-based traversal of taxa, chemicals (CHEBI), environments (ENVO), and phenotypic traits.
- Accelerated study onboarding for 4 disease cohorts (colorectal cancer, lung cancer, epilepsy, skin wounds) by writing scripts that transform BugSigDB differential-abundance exports into per-study directories of taxa lists keyed to NCBI Taxon IDs.
- Reduced manual data-wrangling time by designing a YAML-driven trait-profile generator that parses results across all study directories and emits wide and long TSV tables (study × phenotype) for downstream statistical analysis.
- Improved team reproducibility, as measured by zero environment-setup issues across contributors, by standardizing workflows with Justfile task runners, uv-managed Python environments, and PR-based collaboration on GitHub.

Finance Director — *Various Roles within the ASUC* Sept 2024 – Present

- Administered approximately \$200,000 in student organization funding, as measured by on-time grant disbursements and zero compliance violations, by managing CASSA allocations and awarding grants to student-led proposals across two orgs.

PROJECTS

Canvascope, Lectra & PersonalGraph MCP — canvascope.org Founder / CEO / Lead Developer

- Grew Canvascope to 46 active users across 10+ university Canvas domains by founding a privacy-first AI study assistant (Chrome extension, Manifest V3) with GCP OAuth authentication, local-first indexing, and sub-100ms semantic search.
- Achieved citation-grounded AI responses, as measured by every answer linking to its source document, by designing a RAG pipeline that indexes course PDFs, pages, and assignments into vector embeddings and retrieves relevant context at query time.
- Reduced cloud inference costs and improved user privacy by integrating Chrome’s built-in Gemini Nano for on-device AI and building personalized learning workflows that adapt to a student’s academic background and learning style.
- Enabled cross-device study workflows by developing Lectra (iOS/iPad) with Apple Pencil annotation, DropBridge — a custom high-speed cross-device file-transfer protocol — and a Supabase backend for real-time sync of annotated PDFs.
- Built a sandboxed on-device coding IDE inside Lectra (iOS) — a from-scratch POSIX shell, real git via isomorphic-git bridged to native Swift over a WKWebView, and an embedded Python runtime, all confined to the app sandbox.
- Built PersonalGraph, a TypeScript MCP server that gives AI assistants a persistent personal knowledge graph — 15 tools/resources, a four-tier visibility lattice with secret redaction and sensitive-topic auto-classification, crash-safe atomic writes, and an append-only audit log.

CLENS — github.com/NoelSason/clens | **DataHacks 2026, Scripps Challenge Track** Software Engineer

- Built an iOS app that scores grocery purchases on ocean health by training a binary CoreML classifier on 400+ images to auto-detect receipts vs. product barcodes at 6 fps, using a custom hysteresis state machine to dynamically route the scanning camera UI.
- Generated a live marine stress index by processing Scripps CCE2 mooring datasets over OPeNDAP, applying STL seasonal decomposition and an Isolation Forest to flag anomalies that dynamically penalize high-impact items.
- Enabled on-device footprint scoring by programming a zero-dependency ridge regression model in Python that codegens Swift coefficients to predict CO2, runoff, water, and plastic impact, wired to Apple Vision OCR, a Flask API, and a Supabase database.

TECHNICAL SKILLS & CERTIFICATIONS

Languages:	Python, Java, Swift/SwiftUI, TypeScript, JavaScript, SQL, Bash, HTML5/CSS
ML & Data:	PyTorch, scikit-learn, Pandas, NumPy, DuckDB, MongoDB, ONNX Runtime Web, embeddings & vector search
Quant & Methods:	Linear algebra, statistics & probability, A/B testing, data structures & algorithms
Web & Mobile:	Node.js, Next.js, React, Flask, Fastify, iOS/SwiftUI, Xcode, Chrome Extensions, REST API design
Tools & Platforms:	Git, GitHub Actions, Docker, Linux/Unix, Supabase, Vercel, Stripe, Cloudflare R2, GCP/OAuth
Systems & Homelab:	Raspberry Pi, OpenMediaVault (NAS), Tailscale, Pi-hole, Justfile, uv Cert: AHA BLS